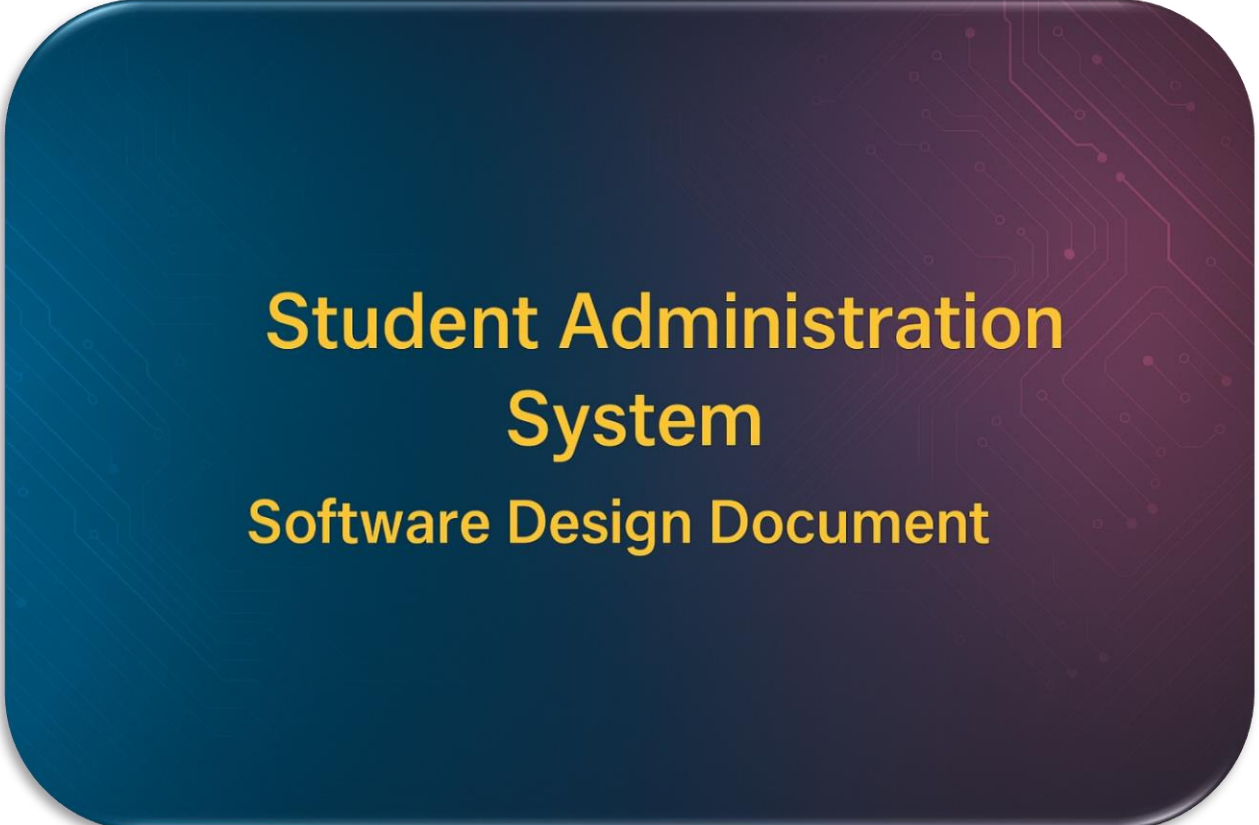


CS241: Software Design and Implementation

Semester 2: Assignment 1 2025



Student Administration System

Software Design Document

Group Number: 14

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Group Members:	Mohammed Afeef	S11229568
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1.0 Introduction and Purpose

Handling student data is one of the major problems that arise in large schools. Lot of paperwork and sometimes even a staff is required to manage data. The system is to be built in order to handle student data like student grades, attendance and class schedule effectively. The system will benefit the schools by decreasing record keeping errors, track students' academic performance, and allow cross communication between parents, teachers, admin staff and students.

This document will outline the design proposal of the Student Administration System (SAS). It is a software solution comprising with complex computing feature that will allow high schools to step towards modernization. This report outlines a detailed guide for everyone involved in the project including the stakeholders, the development team and the end-users to understand the system's features and architecture.

Intended users of SAS includes:

- **Administrators** – The school head/Admin staff/Librarian. They will have access to the admin panel so that they can oversee all the relevant data managed and stored by the system.
- **IT Department** – One of the staffs will be hired by the school to manage the systems functionality and database so that each piece of data is encrypted and is not vulnerable. They will also be responsible for connecting the system to other platforms.
- **Students** – They will have the access to their individual class schedules, assignment/homework module, library module, view grades and update their personal information if required.
- **Teachers** – They will have the access to view students' details, their subject timetable, release assignments/homework, update attendance and fill marksheets.
- **Parents** – They will have the access to view their child's marksheet, attendance/disciplinary reports and academic performance. In case, their child is absent, the system is to notify them.

The structure of Nadi Muslim College, a school with about one thousand-two hundred students was studied to shape the design of SAS. Each year, the number of students enrolling to this high school is gradually increasing leading to a scalable design that is capable of managing and storing large amount of data without a concern. Since the system

will be used by different types of users, the development team enforced a role-based access for all the users so for a better user experience without any flaws in performing their daily tasks. Before the system is released, thorough testing will be done from all four different roles to ensure system integrity. Location of the high school is also taken into consideration so that minimum hinderance occurs during performance.

Design Standards that have been used to ensure that the design meets the school requirements.

- **Spring-boot Architecture** - A straightforward and scalable method for managing categories and items is offered by the Spring-Boot application (Aggarwal, 2023). This architecture also provides to change the project configuration.
- **Role-based access control** – Each user type will have different types of access to the system to ensure information is not accessed by unauthorized individuals and the information remains confidential to others.
- **Normalized database** – This will be done in order to reduce data redundancy and make efficient use of the database tools.

2.0 System Capabilities and Scope

2.1 Stakeholders

Below is the list of stakeholders that are directly or indirectly involved with the Student Administration System (SAS):

Stakeholders	Descriptions
Students	These are individuals who are current registered students at the school and are provided with a Student ID, Student Email and Password (first time login passwords, students can change later).
Teachers	This are currently registered and active instructors at the institution and are provided with an instructor ID, Instructor Email and Password (first time login passwords, teachers can change later).
Parents	These individuals are parents of registered students and only have limited access to the SAS system
Administrative Staff	These are school's admin staffs, and they have the editor's control and control over all major resources and data/information.
IT Department	Manages technical and operational aspects related to the SAS system.

2.2 Functional Requirements

General Stakeholders Functional Requirements

The system shall allow Registrations and Authentication for all stakeholders.

The system shall enforce secure login and role-based access control for all registered stakeholders.

The system shall allow for password reset, stakeholder account recovery and two factor authentication.

Students Functional Requirements

The system shall allow all registered students to register for courses, exams, and electives.

The system shall allow students to access class timetables, schedules, and room details.

The system shall allow students to view their personal profiles and make certain changes based on the read/write access provided by the administrative staff.

The system shall allow students to submit assignments and projects online through allocated drop-boxes.

The system shall notify students regarding upcoming events, assignment/project deadlines, and any changes in classes and schedules.

The system shall allow students to register for extracurricular activities and view their participation records.

The system shall allow students to view their attendance details, exam results and their transcripts.

The system shall allow students to generate their academic and attendance reports.

The system shall provide students with access to all E-learning resources uploaded on the system.

The system shall allow students to search the library book and journal catalogues, view the availability status and make a reservation, while also ensuring that the reserve books are marked as unavailable for other users.

Teacher Functional Requirements

The system shall allow teachers to record, monitor and modify student attendance.

The system shall allow teachers to record, monitor and update student disciplinary log.

The system shall allow teachers to record and update grades received in exams, assignments, and projects.

The system shall allow teachers to upload course materials, and other e-learning resources.

The system shall allow teachers to generate student performance reports using the student grades and share reports and feedback with students and parents.

The system shall provide teachers with access to class schedules and timetables.

The system shall allow teachers to create, publish and update announcements and reminders for students and parents.

The system shall allow teachers to manage and track extracurricular activities.

Parents Functional requirements

The system shall allow parents to have read-only access to their child's academic records, extracurricular activities records and report card.

The system shall allow parents to view announcements, events, schedules together with notices regarding parent-teachers meeting and deadlines.

The system shall allow parents to view their child's attendance and disciplinary records.

Administrative Staff Functional Requirements

The system shall allow administrators to manage new or existing student admission, registration, and student records.

The system shall allow administrators to assign teachers to classes and allocate rooms for class sessions.

The system shall allow administrators to create and publish class timetables and schedules for exams.

The system shall provide administrators access to publish notices, newsletters, and announcements.

The system shall allow administrator to view and manage all student disciplinary records, that are recorded by the teachers.

The system shall allow administrators to view and manage all student grades.

The system shall provide administrators access to generate student academic report.

IT Department Functional Requirements

The system shall allow IT staff to manage user account registration, role-based access, and permission.

The system shall allow IT staff to monitor the Student Administration system performance and generate reports.

2.3 Non-Functional Requirements

Performance -The system to accommodate many concurrent users without any performance degradation.

- The system shall respond to any user's requests within 3 seconds and provide real-time updates.

Scalability: The system should be able to manage a large number of users and transactions during peak times (like annual reports).

- As the number of users and records rises, the system must be scalable enough to accommodate growing data processing and storage capacities.

User Experience: The system shall have a user-friendly intuitive interface that is suitable for the users

- After a maximum of one week of training, users will be able to operate the system.
- Mobile devices with internet access will be able to simply access the system.
- The system must be capable of processing data effectively and promptly delivering appropriate responses.

Data Integrity: The system shall backup all the data in case of a downtime.

- The system must have a copyright to prevent unauthorized use or duplication without authorization.

Security and Privacy: Strict adherence to data privacy regulations and high standards of data protection

- The system will adhere to the guidelines established by the Ministry of Education concerning the safeguarding of students' national education data.
- There must be official documentation for the system that covers its security, dependability, safety, absence of malicious activity, and risk management protocols.
- The system will guarantee that all of its data is kept private and secure by implementing a privacy policy.

Conflicting Non-Functional Requirements

Performance vs. Scalability: As the system expands, performance can deteriorate. It might slow down at times of high load (end-of-term). Finding a balance between responsiveness and load-handling capacity is one such challenge.

Complex Computing Features

Data Processing – The system can do real-time data processing for example, if the teacher uploads an assignment for a particular class, it will be available on their assignment module.

Scalability – Hosting data on cloud enables the system to handle the growth rate of users using the system through auto-scaling and multi-tenancy.

Backup and Disaster Recovery – The system backs up data on daily basis during nighttime so that the performance is not affected and prevents data loss.

Cross-platform Access – The use of responsive design of UI by the developers makes the service available 24/7 on mobile, tablets, and desktop devices through web browsers required that the user has internet access.

Automation – The system is able to identify the students that need to improve on their academics through performance tracking. This helps teachers to provide some strategies for the students so that they can improve on their academic performances.

Subsystem Integration - Auto generate timetable. For instance, if the admin sets math class for year 11B at 8am. The system automatically fills in the class timetable for the mathematics teacher that he has a class at 8am for year 11B, which is shown in his timetable.

Role-Based Access – All the different types of users will have their own user panels which prevents unauthorized access, provides a smooth workflow environment and ensures privacy.

3.0 Risk Analysis

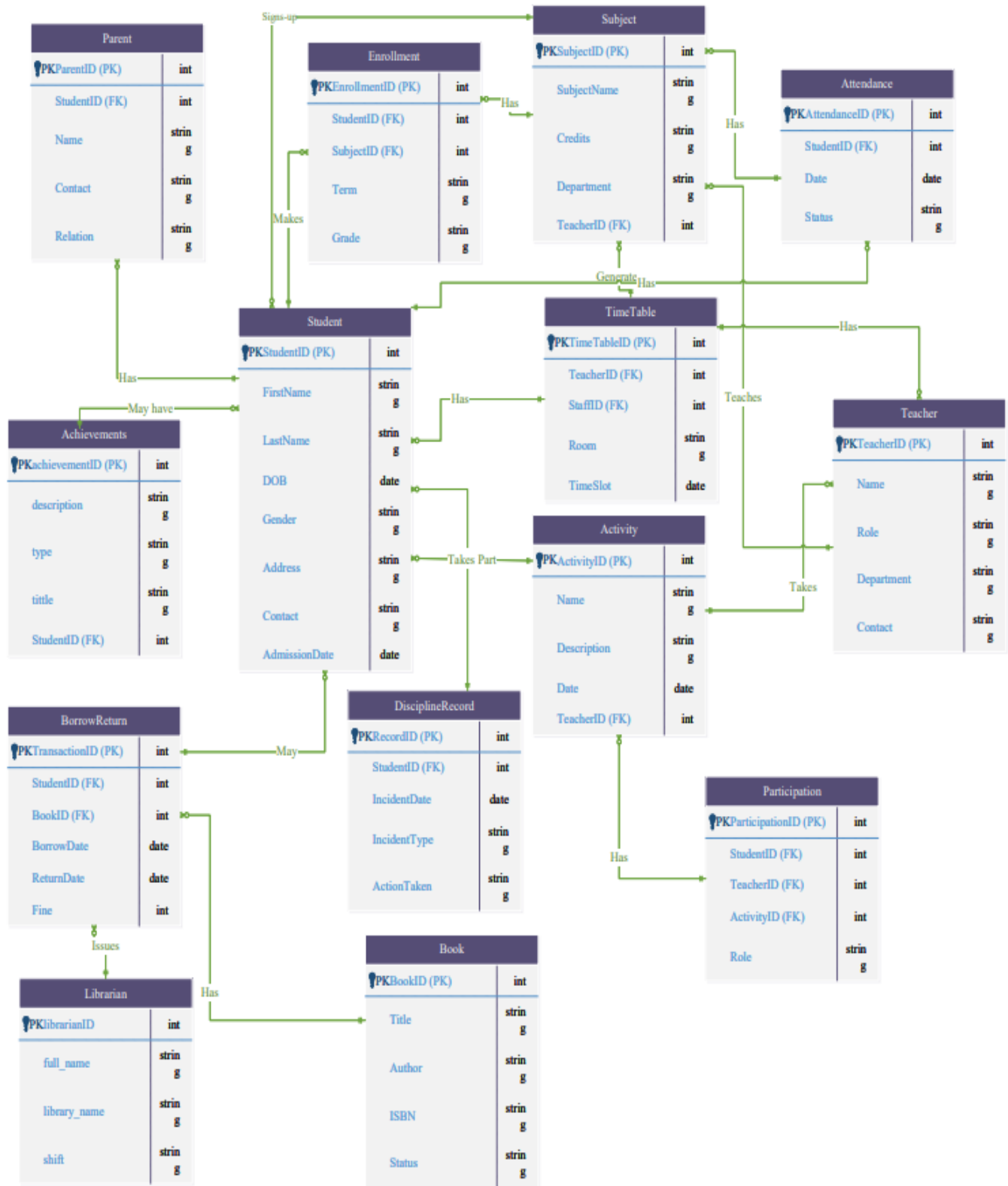
For each source of risks causing potential loss to a software project, the exposure is defined as the product of potential loss, multiply by the potential loss. (Boehm, 1989)

Risk Area	Description	Likelihood	Impact
Data Privacy	Unauthorized access to sensitive student/staff information that can lead to identify theft.	High	Critical
Data Security	Poor encryption of data. May lead to hacking, theft or unauthorized changes.	Medium	Critical
User Authentication & Access	Weak access control may allow unauthorized users to have access to sensitive data. This includes poor password or role-based permission.	Low	High
Data Integrity	Database design not being compatible with the school system. May lead to student data loss from the system.	Medium	Catastrophic
Disaster recovery	An inefficient backup system has risk of permanent data loss from disasters or sometime accidental deletion.	High	Catastrophic
Availability	Network outages may interrupt system functionality, affect access to data and disrupt administration. This can further lead to increased risk of breaches.	High	Catastrophic
Scalability	Failure of the system to adapt to an increasing number of students in the school. The system may go down due to heavy traffic of users.	Medium	Critical

The best way to mitigate risk is through TARA method, which involves Transferring, Accepting, Reducing and Acknowledging risks. An effective TARA method will provide efficient and effective risk mitigation strategy and reduce risk exposure.

Risk Area	Mitigation
Data Privacy	Use of role-based access might reduce the risk since users will be able to perform tasks based on their roles.
Data Security	Data will be encrypted for stored records. Firewall deployment to detect virus in the system.
User authentication & Access	Team has decided that role-based access will be given to all users to help protect their data. Each user will have different types of access for e.g. Students will only have students' access.
Data integrity	Use of version control for configuration and scripts.
Disaster Recovery	Data will be backed up daily on cloud to avoid data loss. Preparation of data recovery plan with defined recovery time objectives.
Availability	Use of load balancing during high traffic. Maintenance to be scheduled in nighttime.
Scalability	The use of spring boot architecture, a scalable system will be designed which can handle the increased rate of enrollment for students.

4.0 Data Modeling: Entity Relationship Diagram



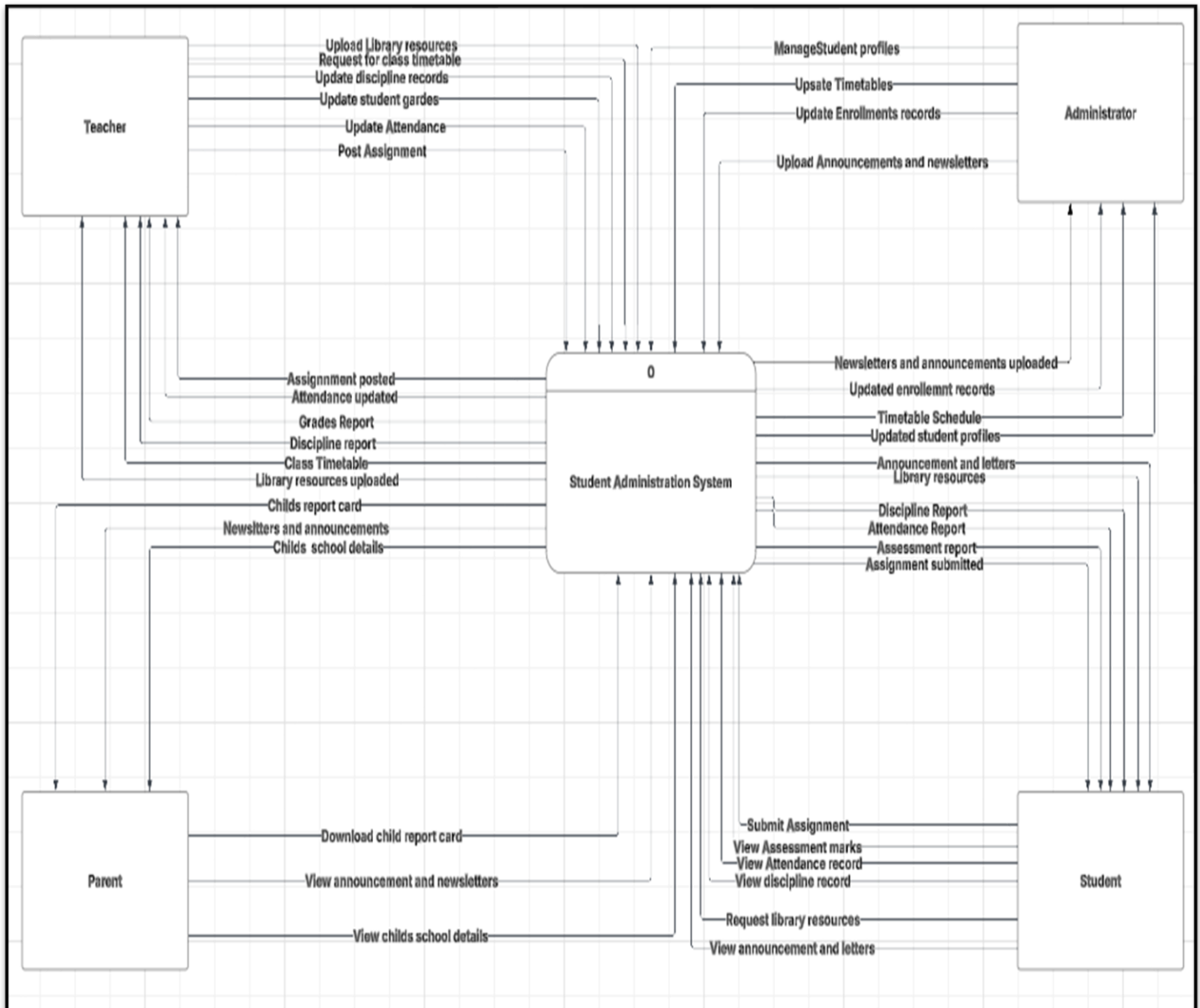
5.0 Application Architecture

5.1. Structured design artifacts

5.1.1. Structured Chart

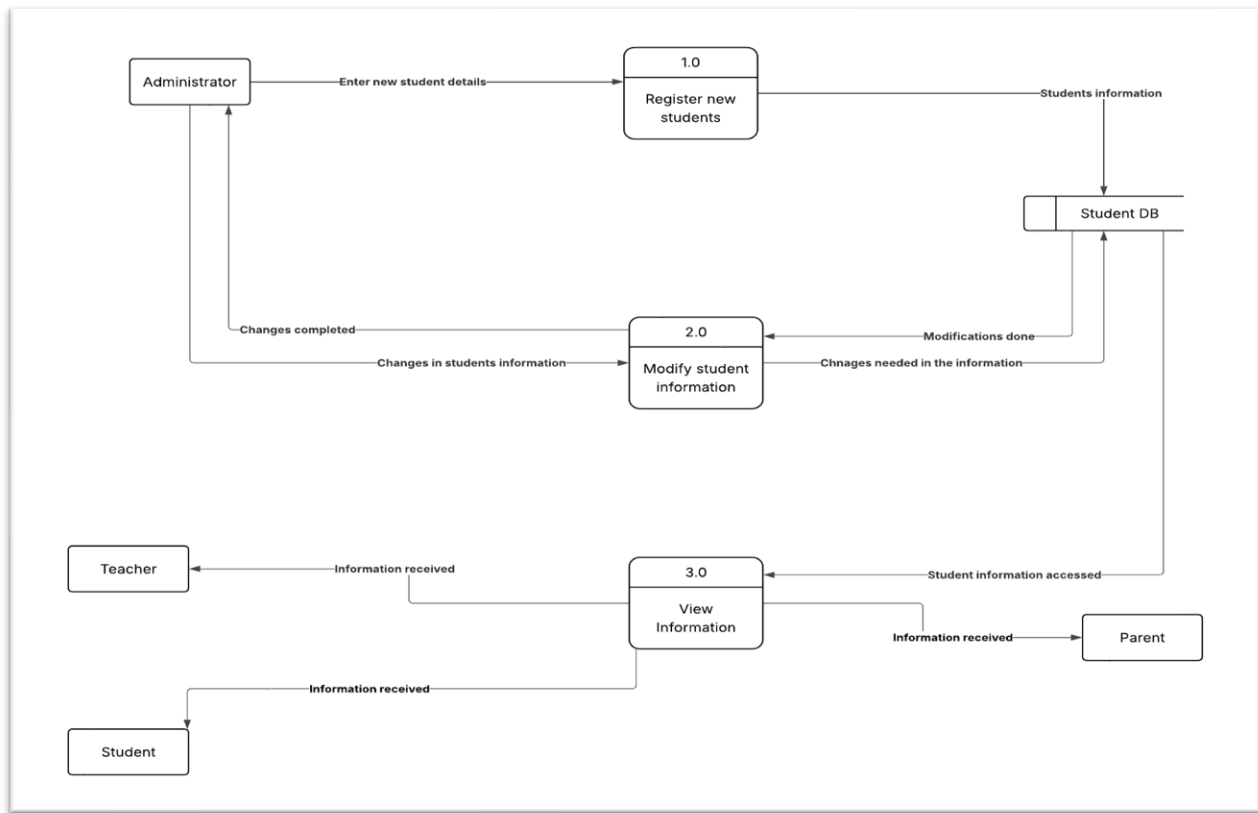


5.1.2 System Context Diagram/DFD Level 0 Diagram

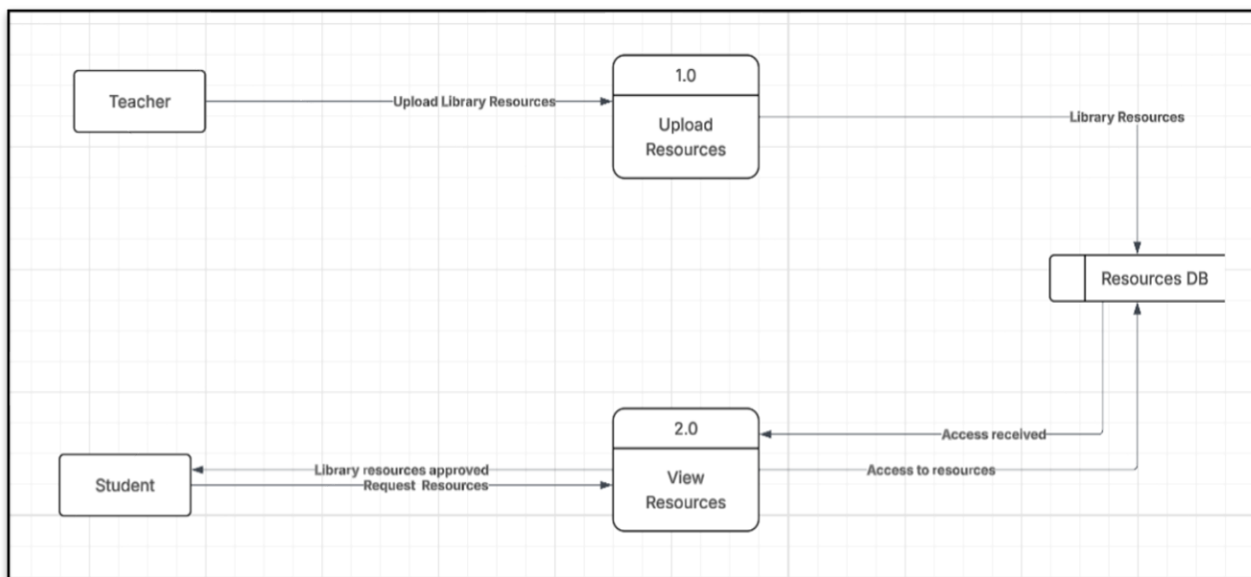


Level 0 Data Flow Diagram

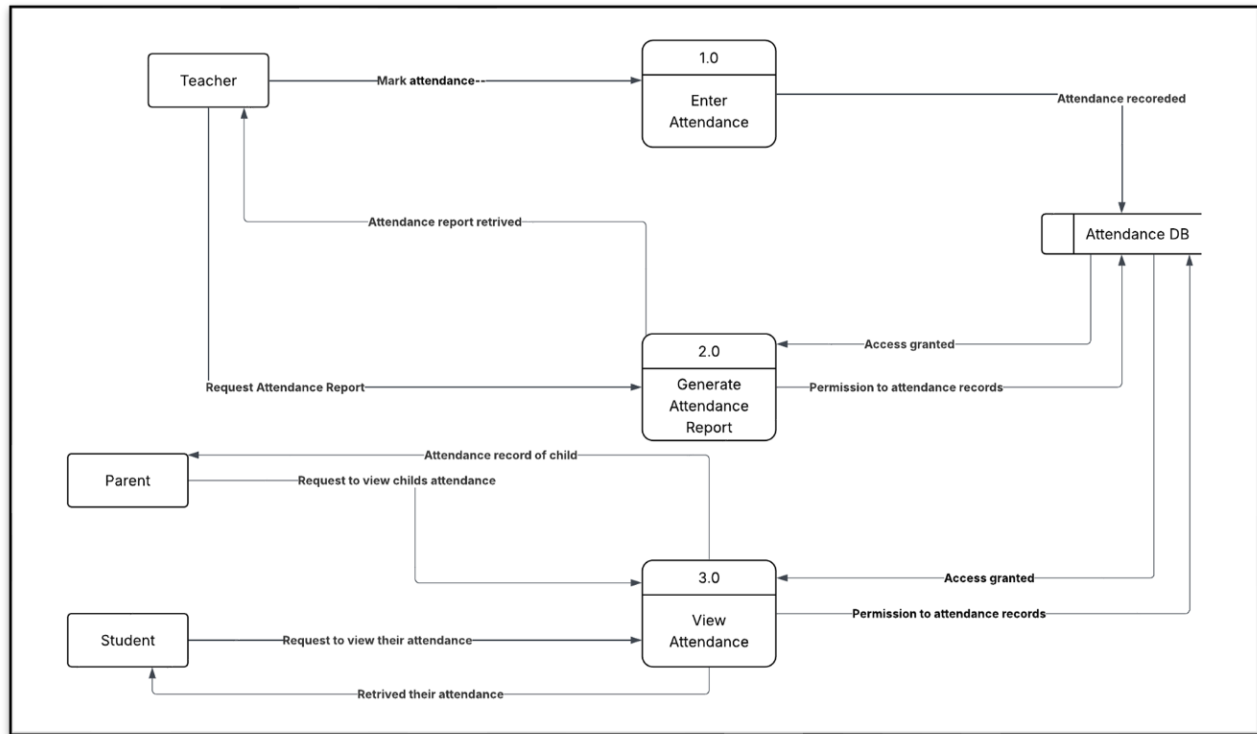
Student Information Repository Module



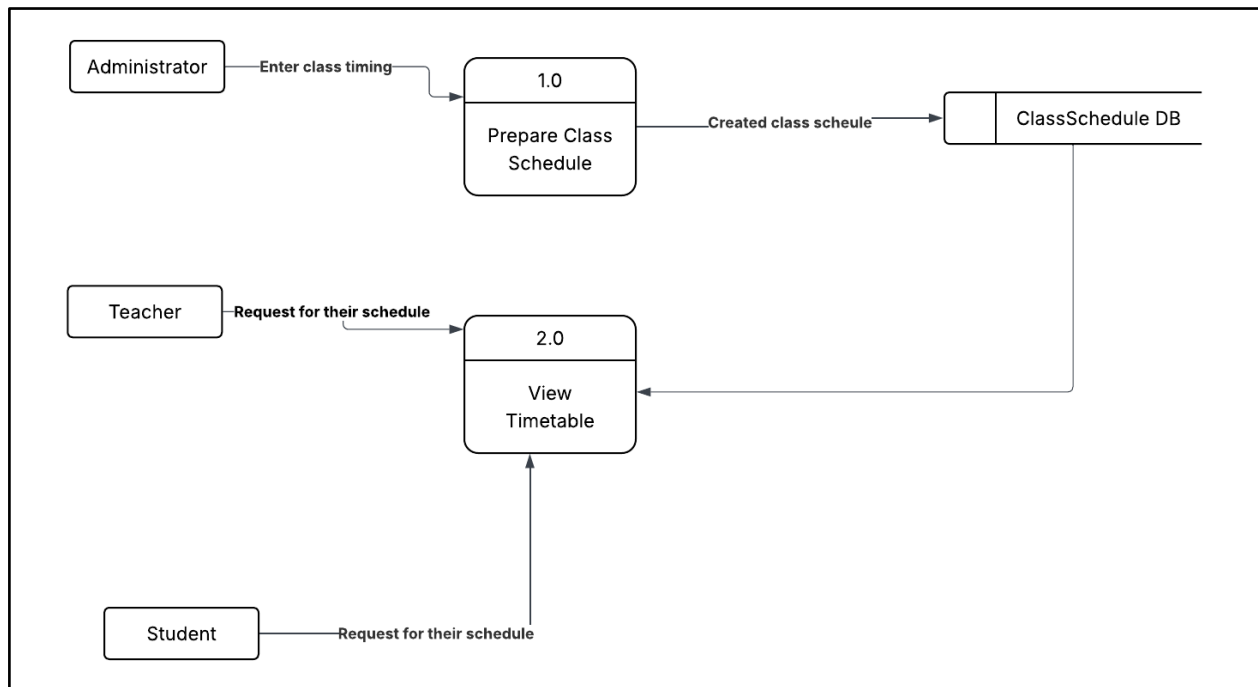
Library Resource Management Module



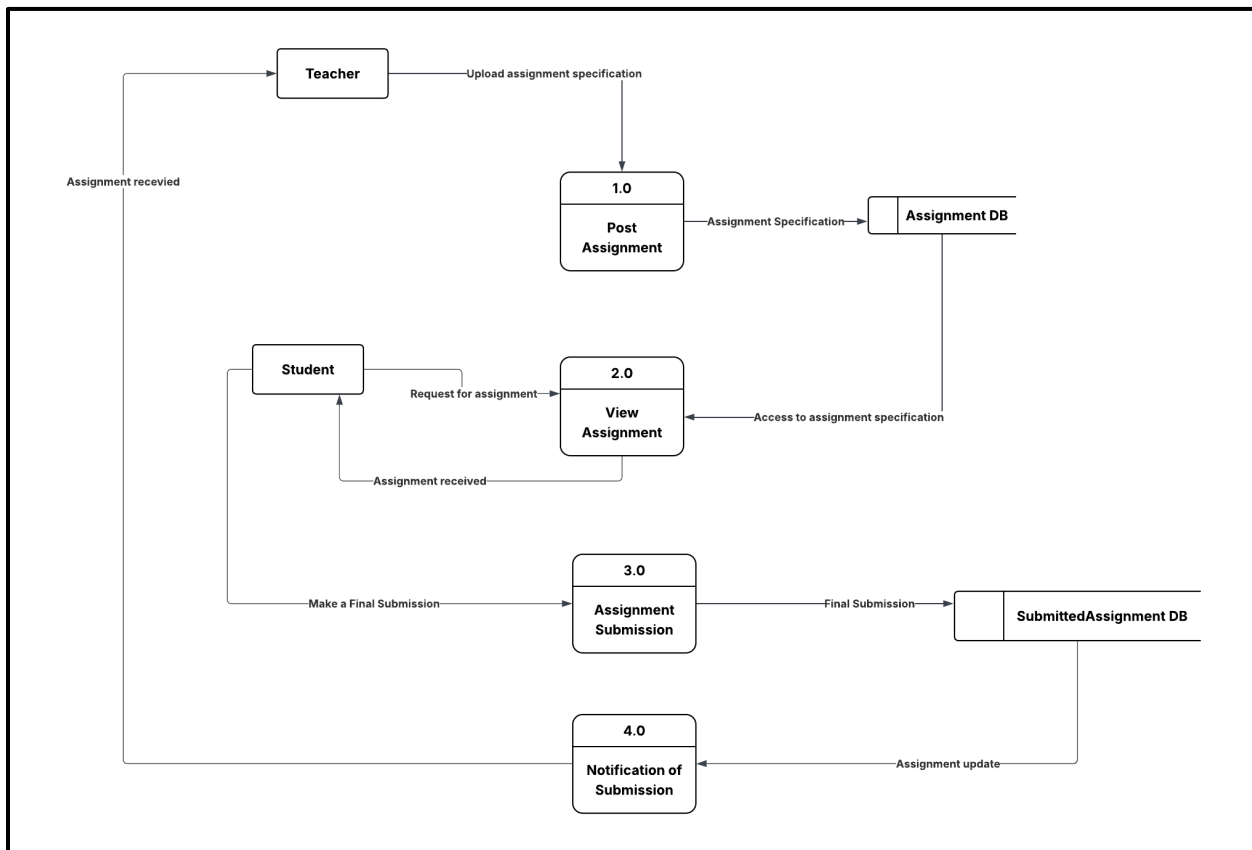
Attendance Tracking Management Module



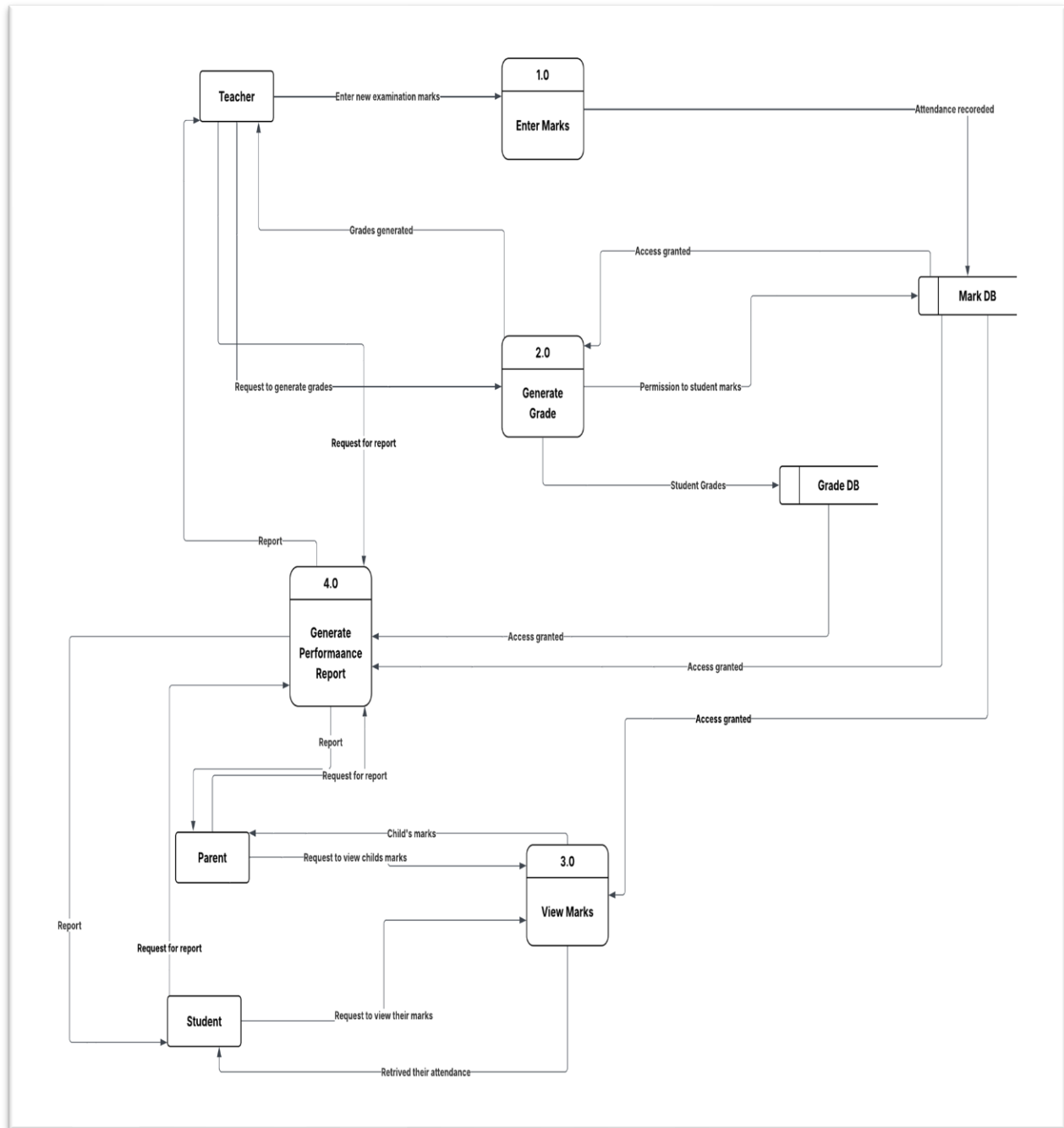
Timetable and Scheduling Management Module



Assignment Management Module

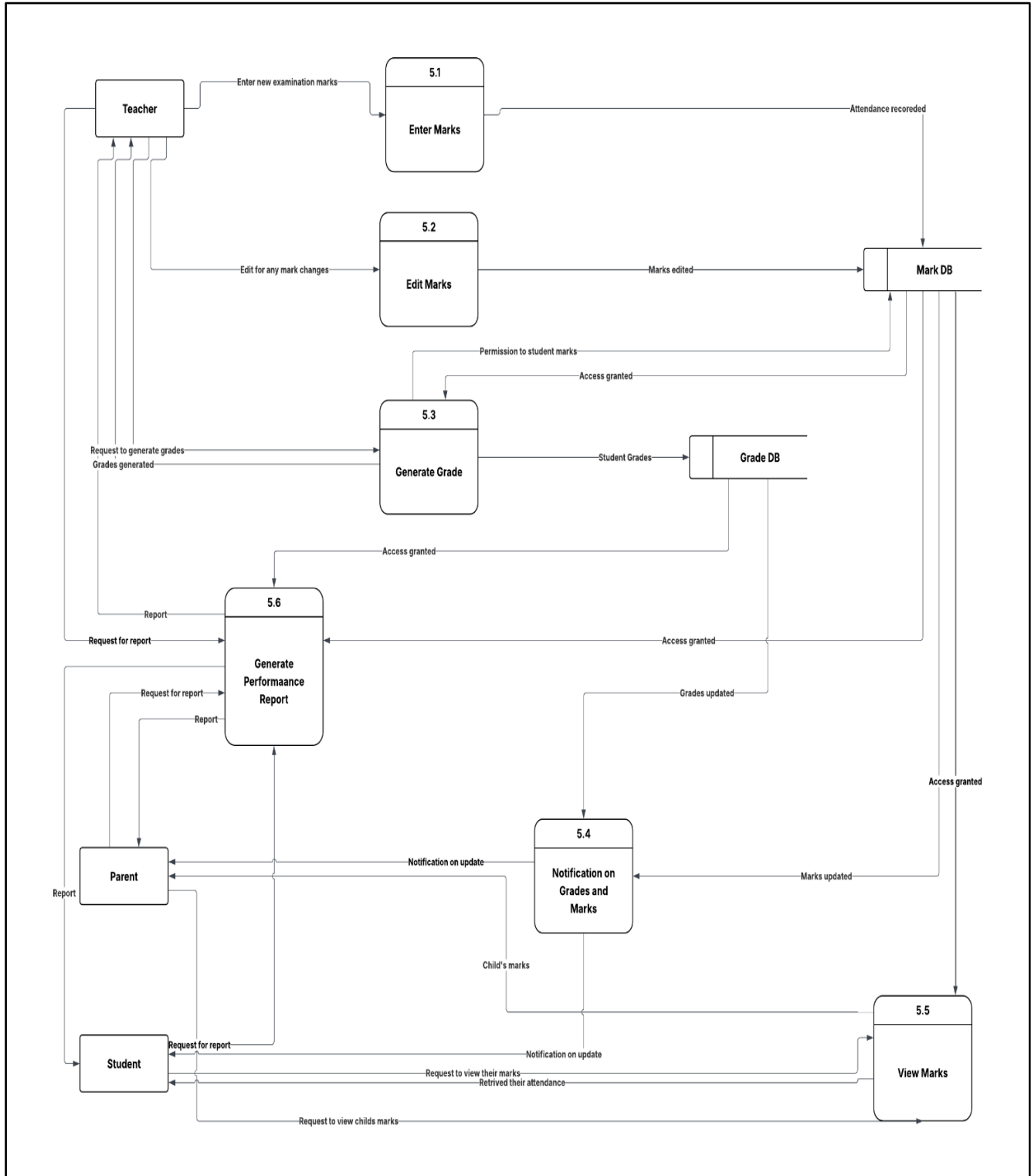


Student Grade Management System Module



5.1.2.1 Primitive DFD for a Major Module

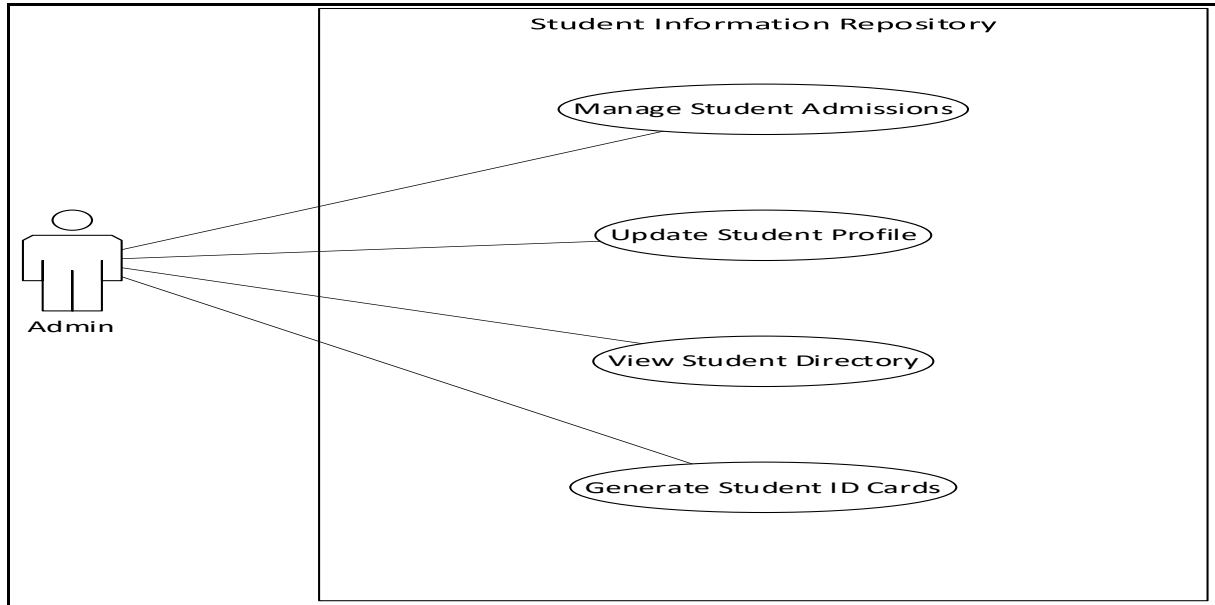
Student Grade Management System



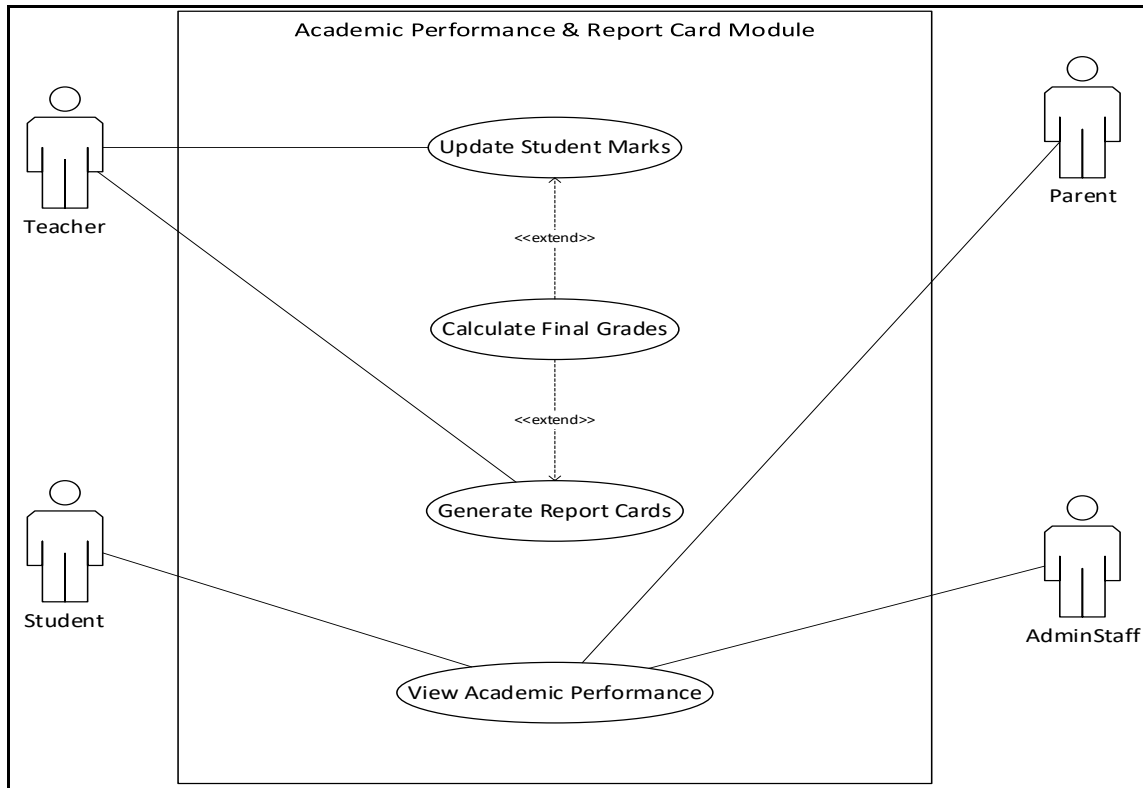
5.2 Object Oriented Design Artifacts

5.2.1 Use-Case Diagram and Description

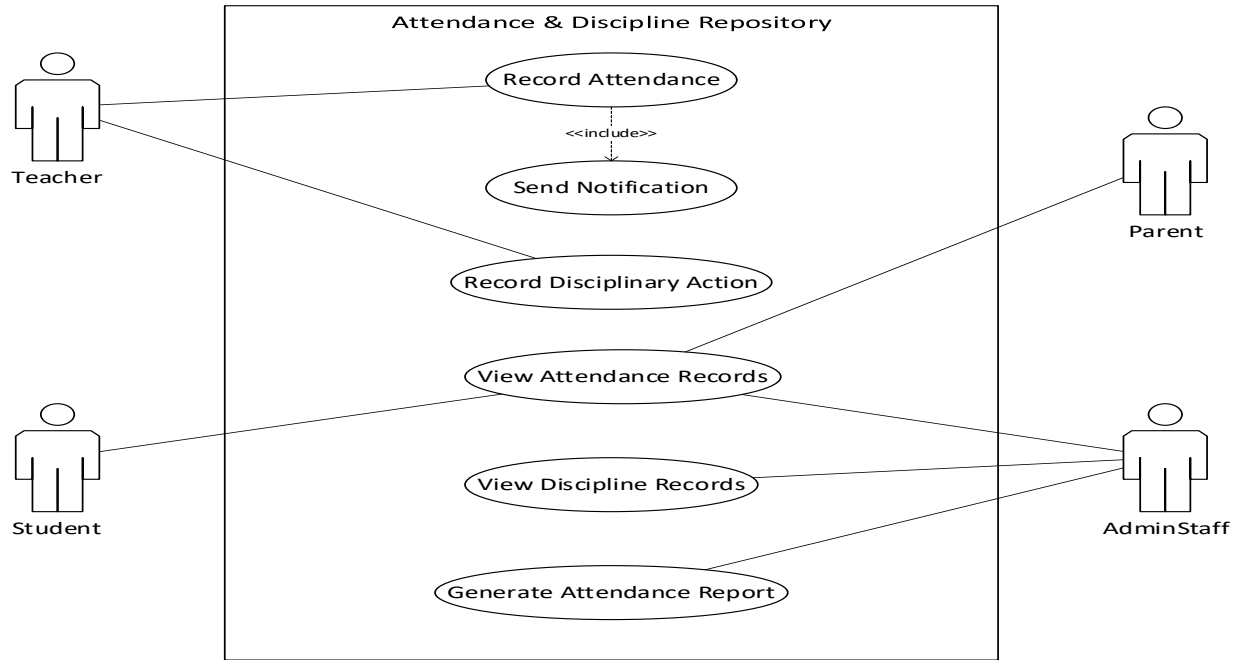
Student Information Repository



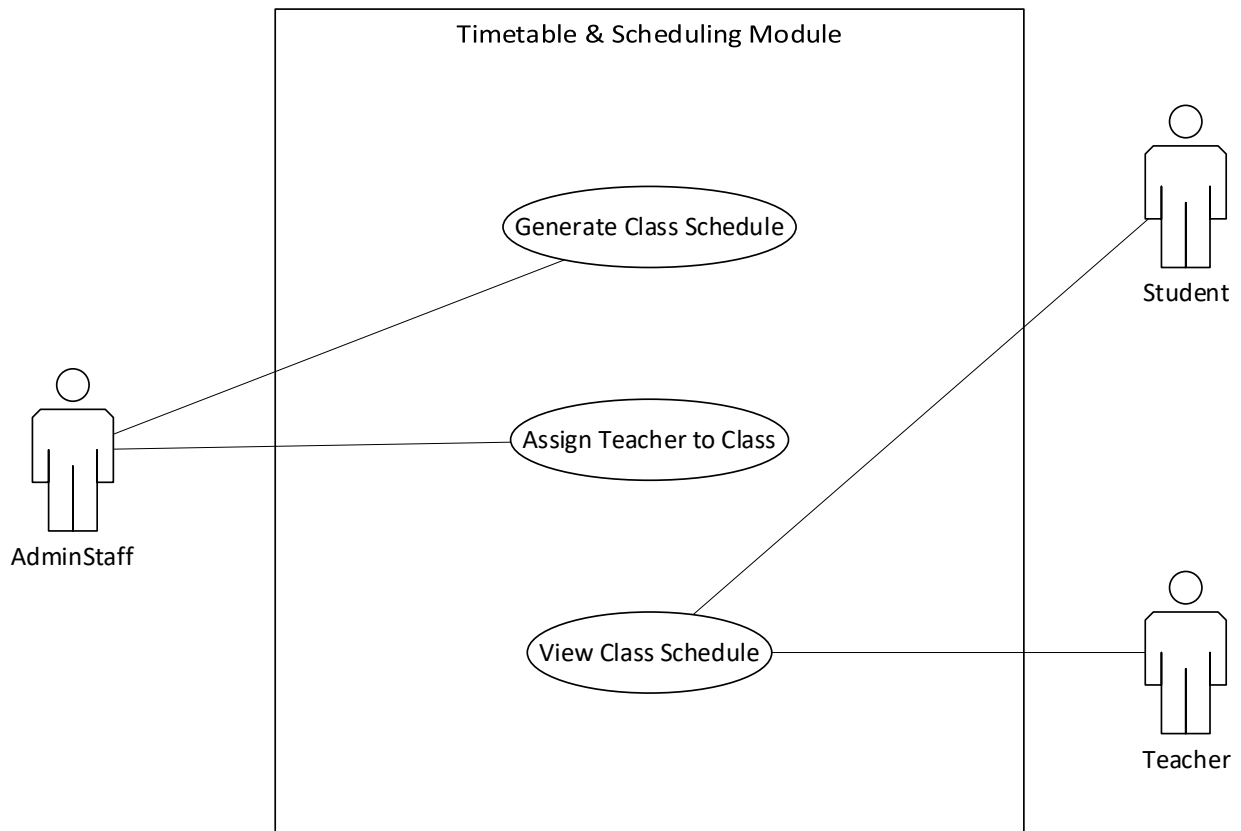
Academic Performance & Report Card Module



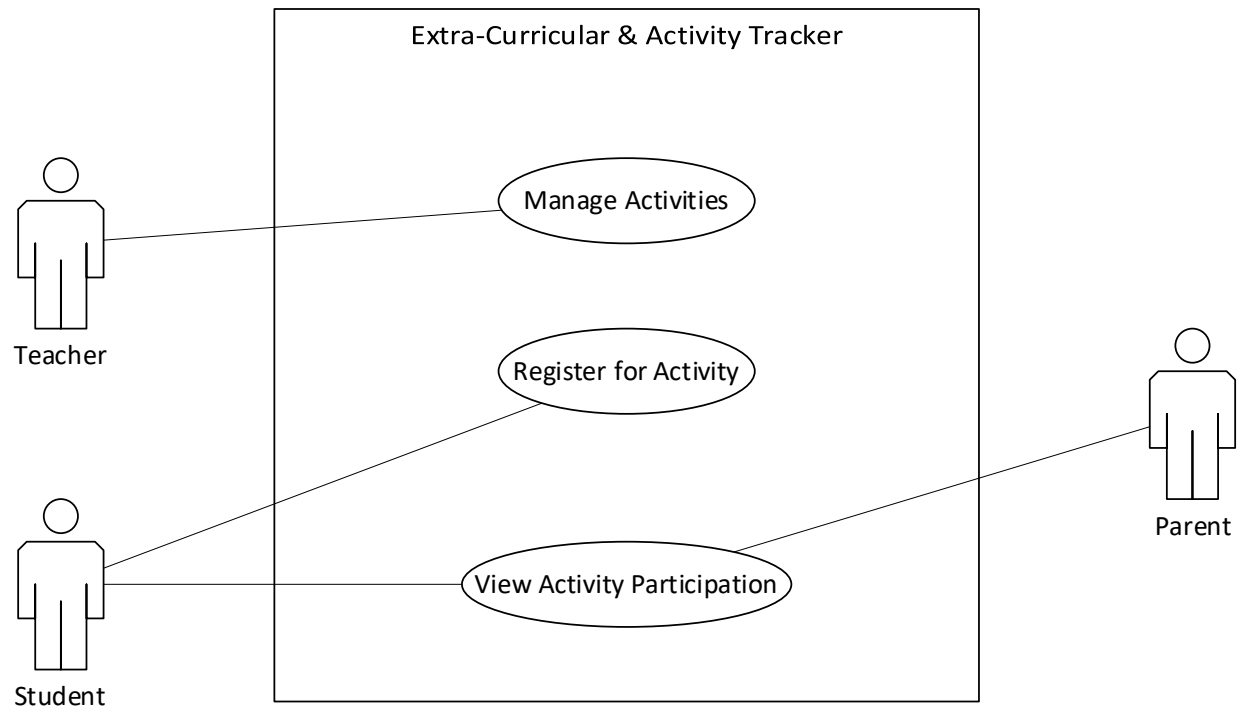
Attendance & Discipline Repository



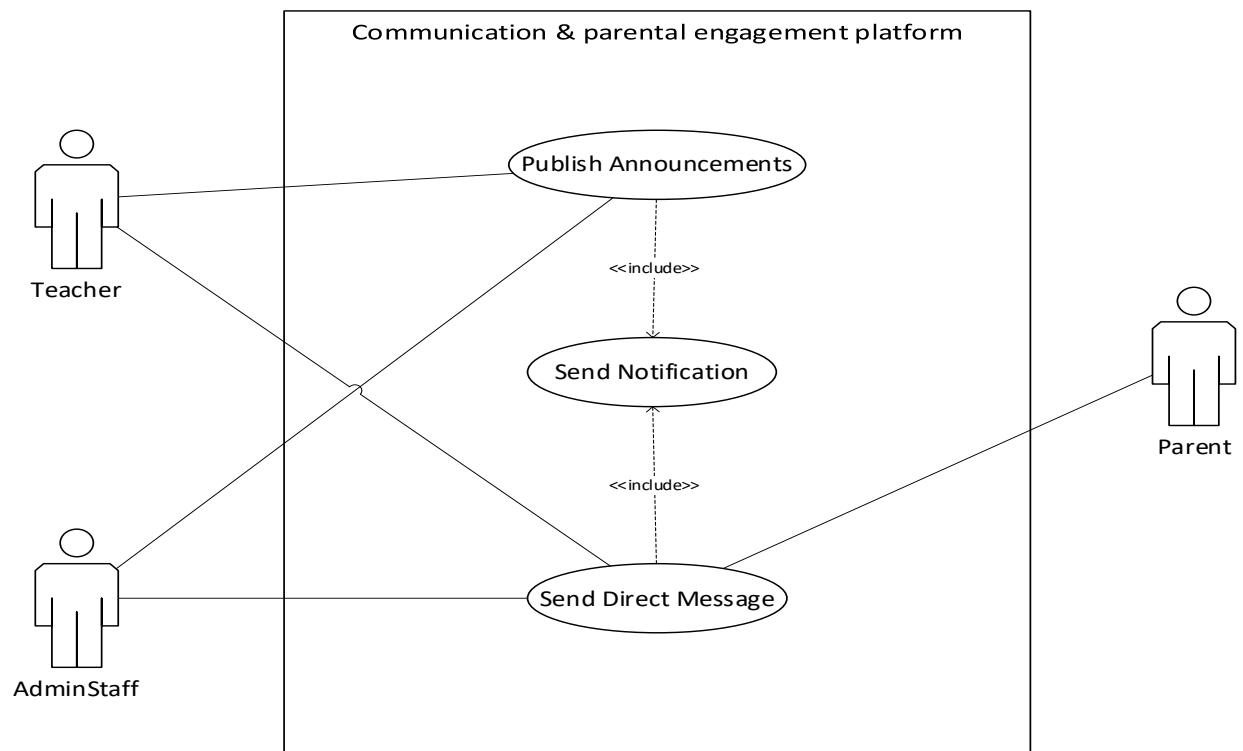
Timetable & Scheduling Module



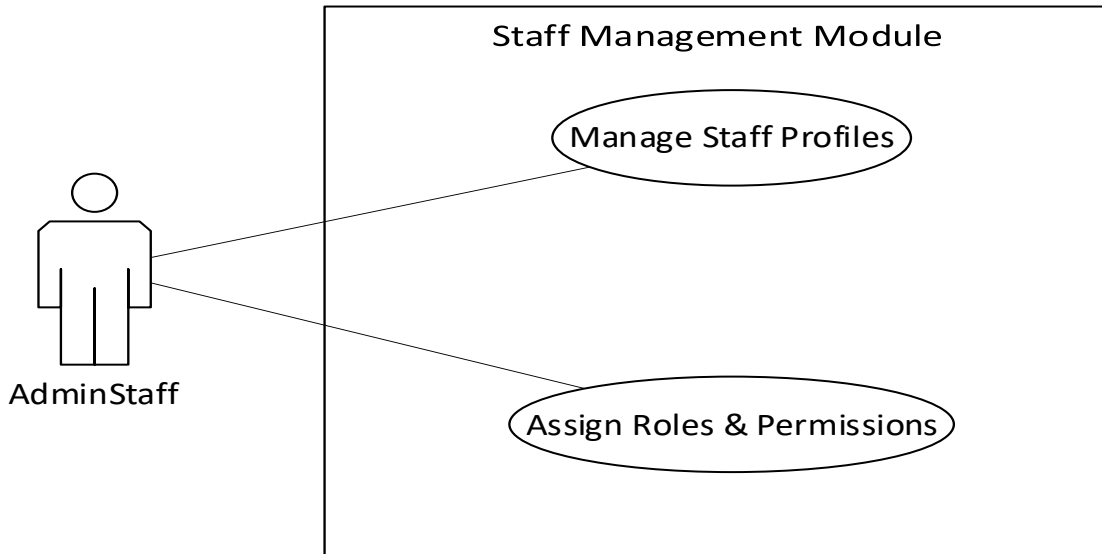
Extra-Curricular & Activity Tracker



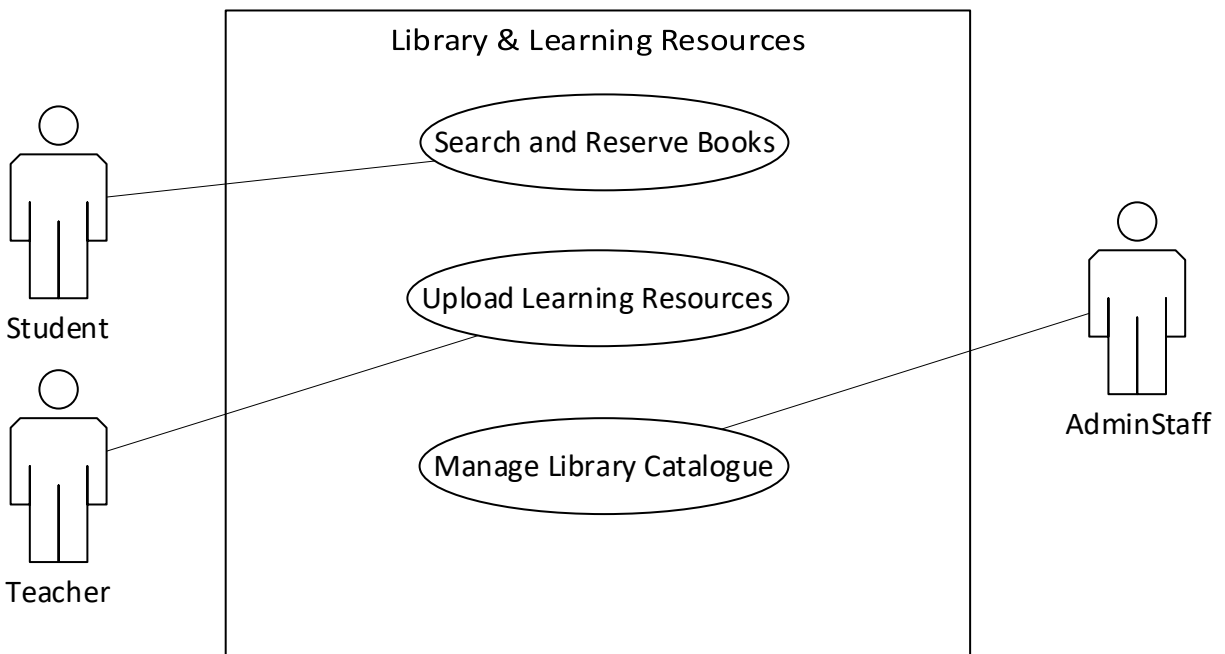
Communication & Parental Engagement Platform



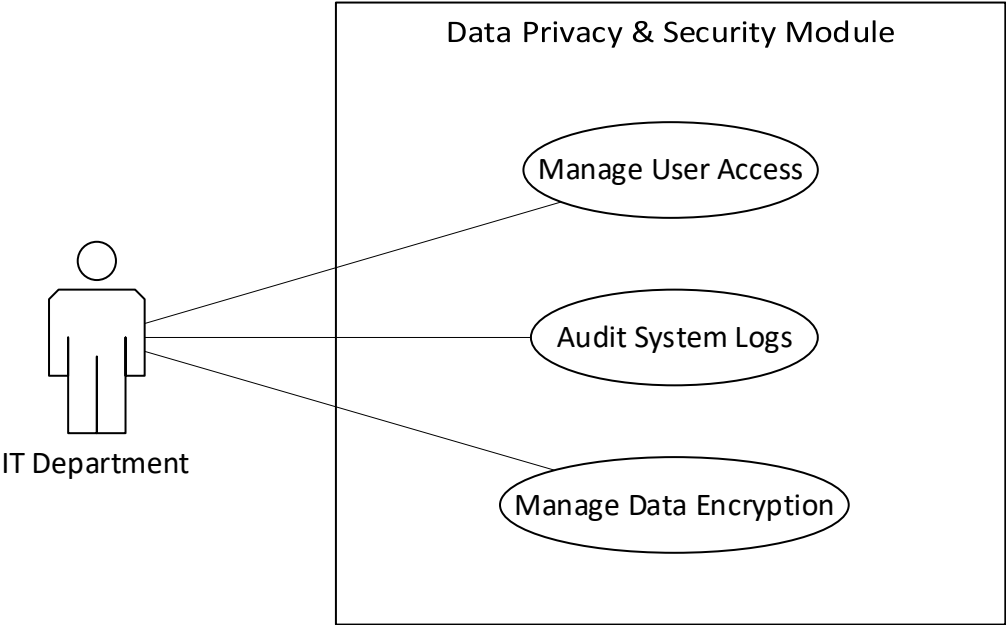
Staff Management Module



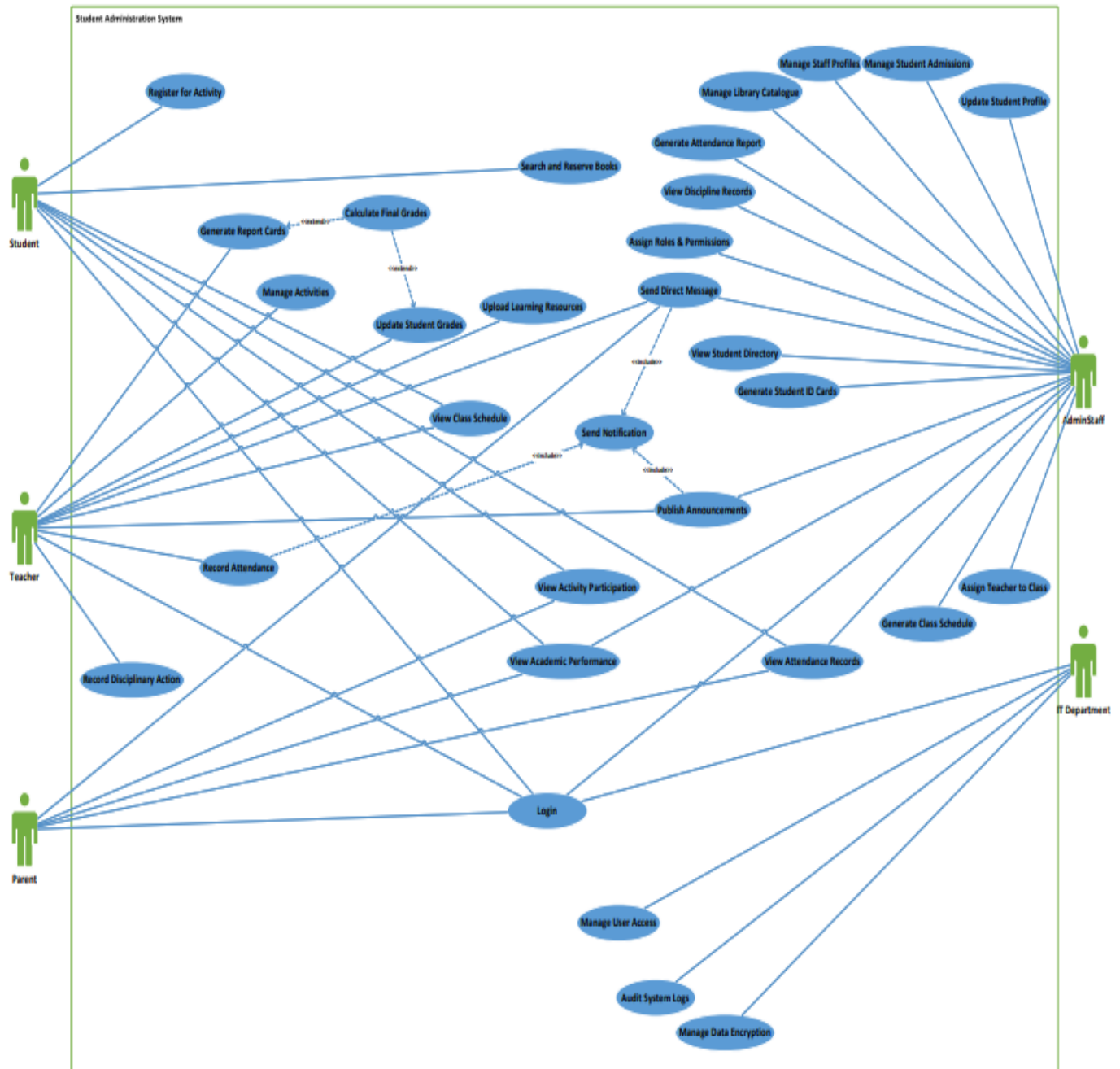
Library & Learning Resources



Data Privacy & Security Module



Overall Use Case Model

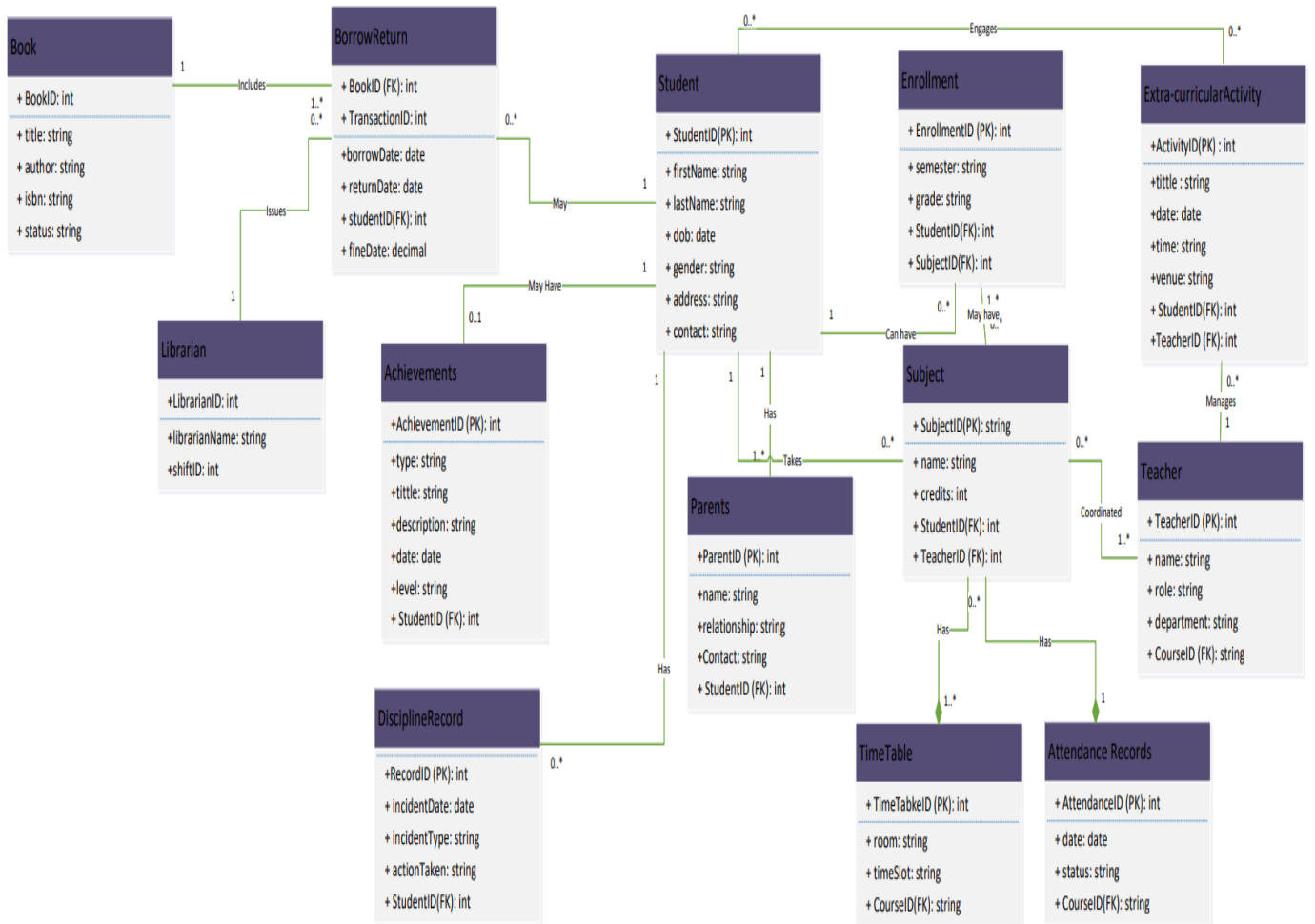


A Major Use Case Description

Use case Name	Search and Reserve Books	
Scenario	A student searches for book in the library catalogue and reserves an available copy.	
Triggering Event	The student needs to borrow a book of their interest.	
Brief Description	This particular use case allows students to search for a book in the digital library, check its availability, and place a reservation on it so that you can pickup later.	
Actors	Student	
Preconditions	The student must be logged in. The library catalogue system should be operational.	
Postconditions	The status of the book is updated to “Reserved”, a reservation is then created, and the student receives a notification of confirmation.	
Flow of Activities	Actor	System
	1. Student logs in to the SAS system.	1.1 System verifies the login credentials.
	2. Student navigates to the library section.	2.1 The system displays the library search interface.
	3. Student enters the search criteria (author of the book, title of the book, etc.) and clicks on the “Search” button.	3.1 The system queries the catalogue database and returns a list of matching books with their availability status.

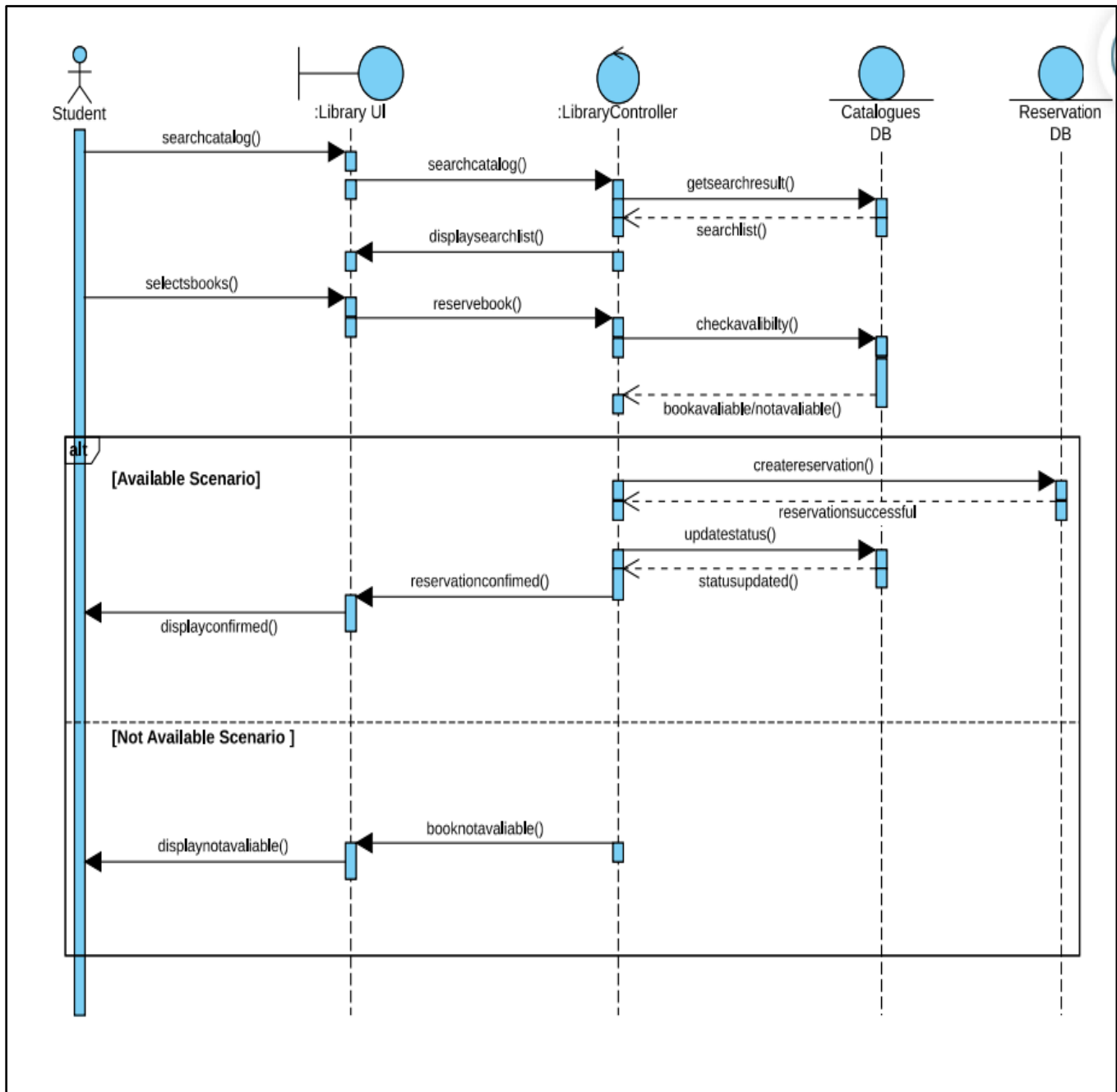
	4. Student selects an available book from the list.	4.1 The system displays a detailed book information and a “Reserve” button on the side.
	5. Student clicks on the “Reserve” button.	5.1 The system verifies the student’s account to see whether its in good standing (no previous blocks).
		5.2 The system creates a reservation record in the database.
		5.3 The system updates the book’s status as “Reserved”.
		5.4 The system displays a confirmation message to the student about the successful reservation.
Exception Conditions	➤ Failure to login due to incorrect credentials submitted. ➤ No search results found for the student’s query. ➤ The book is not available. (already checked out or reserved by another student). ➤ The student accounts are not in a good standing (e.g. the account maybe blocked or overdue books).	

5.2.2 Domain Model Class Diagram

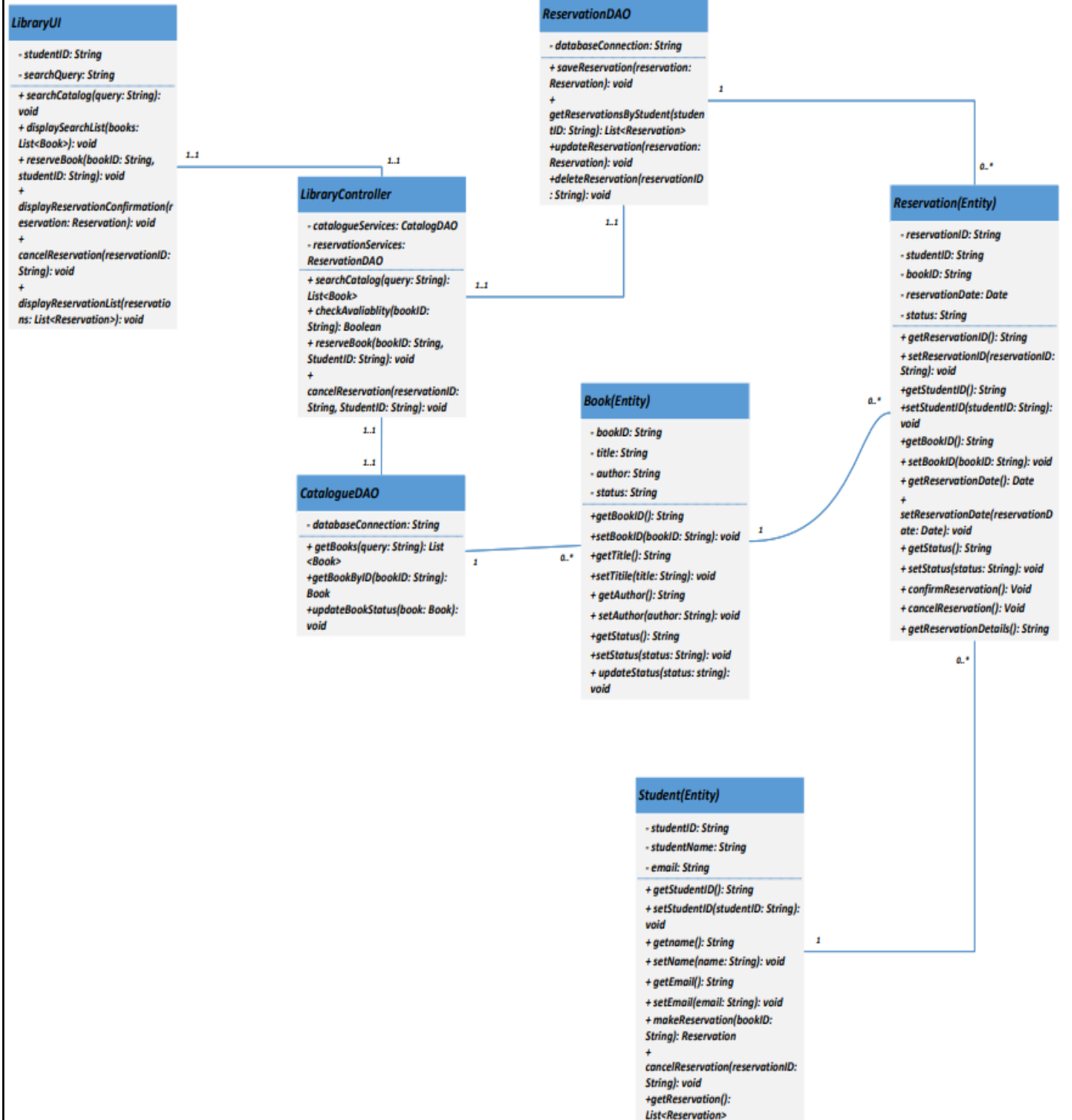


5.2.3 A Major Sequence Diagram

Search and Reserve Books



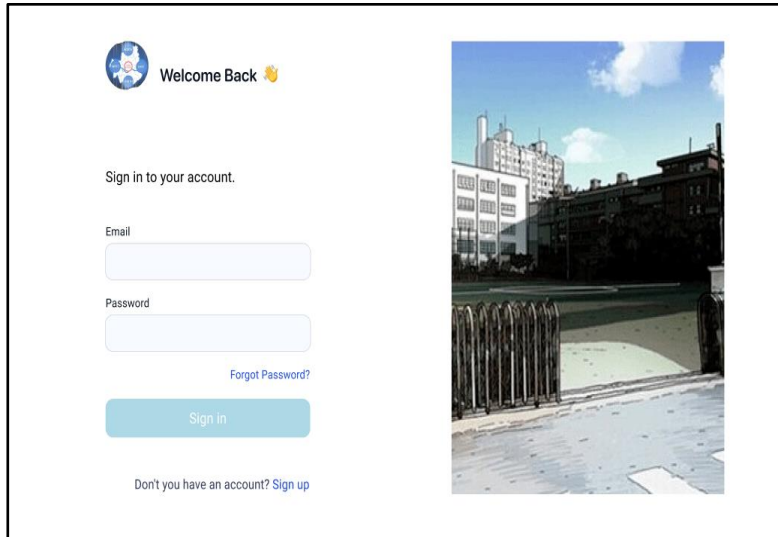
5.2.4 Design Class Diagram



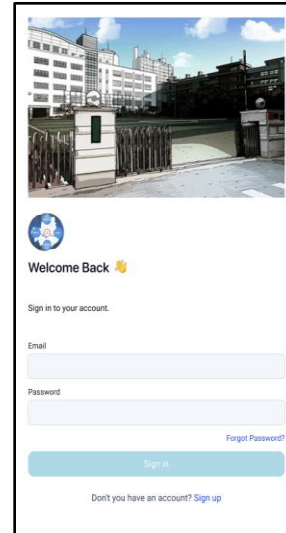
6.0 User Interfaces

Login Page

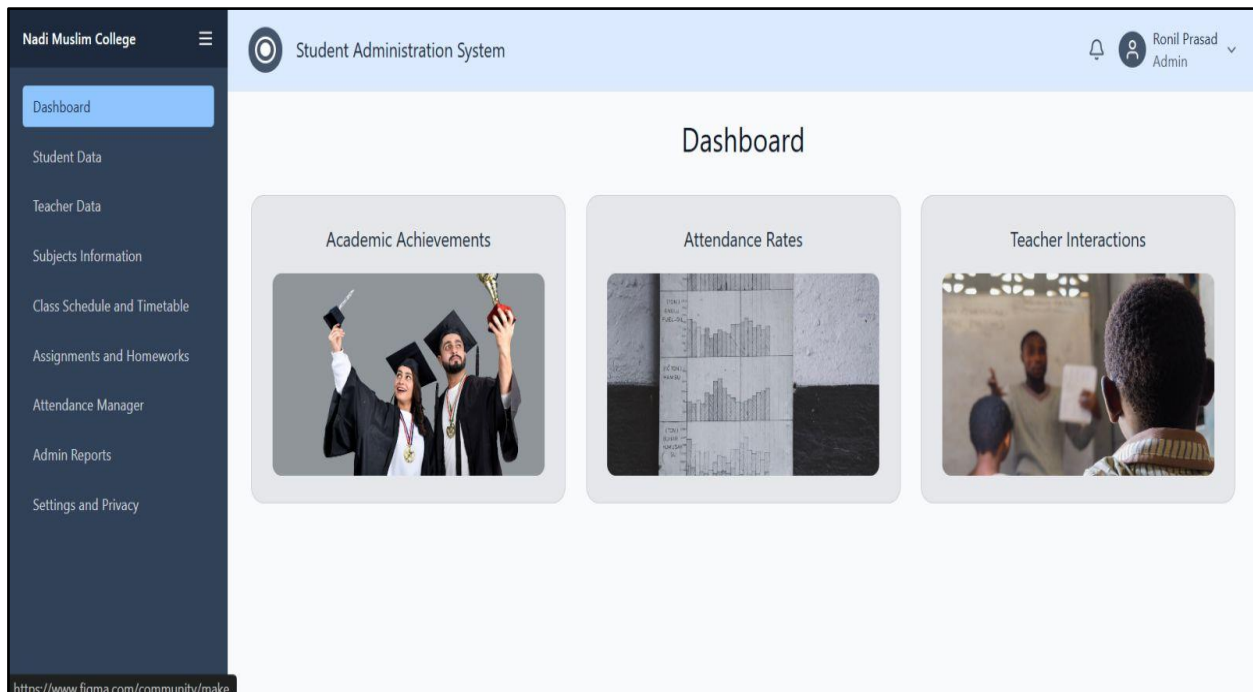
Desktop view



Mobile View



Admin Dashboard



Student Data

Student Data

Teacher Data

Subjects Information

Class Schedule and Timetable

Assignments and Homeworks

Attendance Manager

Admin Reports

Settings and Privacy

Student Data

← Back to Student List

Profile

Enrollment History

Grades

Attendance

Behaviour Record

Parents/Guardian



Student ID: S102021

Name: Daniel Park

DOB: 11/06/2008

Gender: Male

Citizenship: Fiji

Edit

Active Enrollment:

Year: 11

Stream: B

Class Teacher: Mr Rajesh Patel

Subjects	Teacher
Mathematics	Mr Rajesh Prasad
English	Ms Anita Kumar
Biology	Mrs Priya Devi

Student Grades

Nadi Muslim College

Dashboard

Student Data

Teacher Data

Subjects Information

Class Schedule and Timetable

Assignments and Homeworks

Attendance Manager

Admin Reports

Settings and Privacy

Student Administration System

Ronil Prasad Admin

Student Data

← Back to Student List

Profile

Enrollment History

Grades

Attendance

Behaviour Record

Parents/Guardian

Grades

Mathematics: A

English: B+

Biology: A-

Physics: B

Chemistry: A

Teacher details

Nadi Muslim College

Dashboard

Student Data

Teacher Data

Subjects Information

Class Schedule and Timetable

Assignments and Homeworks

Attendance Manager

Admin Reports

Settings and Privacy

Student Administration System

Teacher Data

Total Teachers

37

Select Subject

-

Filtered Results

37

Search Teachers:

Enter teacher name, ID, or subject...

Subject: All subjects

+ Add New Teacher

Teachers

37 teachers found

Teacher ID	Name	Subject	Contact	Email	Actions
T101001	 Rajesh Prasad	Mathematics	9871001	rajesh.prasad@jwhs.ec	 View  Edit  Del
T101002	 Anita Kumar	Mathematics	9871002	anita.kumar@jwhs.edu	 View  Edit  Del
T101003	 Sunil Chand	Mathematics	9871003	sunil.chand@jwhs.edu	 View  Edit  Del
T101004	 Kavita Devi	Mathematics	9871004	kavita.devi@jwhs.edu	 View  Edit  Del

Attendance Module

Nadi Muslim College

Dashboard

Student Data

Teacher Data

Subjects Information

Class Schedule and Timetable

Assignments and Homeworks

Attendance Manager

Admin Reports

Settings and Privacy

Student Administration System

Attendance Manager

Record Attendance

View History

Generate Report

Date: 11/08/25

Time: 09:01

Search student...

ID	Name	Present	Absent	Late	Excused
S101234	Anil Prasad				
S101235	Kamla Devi				
S101236	Rohit Singh				
S101237	Priya Nandan				
S101238	Sunil Kumar				
S101239	Asha Rani				

Attendance Summary

Total Present:

12

Total Absent:

2

Total Late:

0

Total Excused:

1

Total Enrolled:

15

Attendance Rate

80%

7.0 Implementation Plan

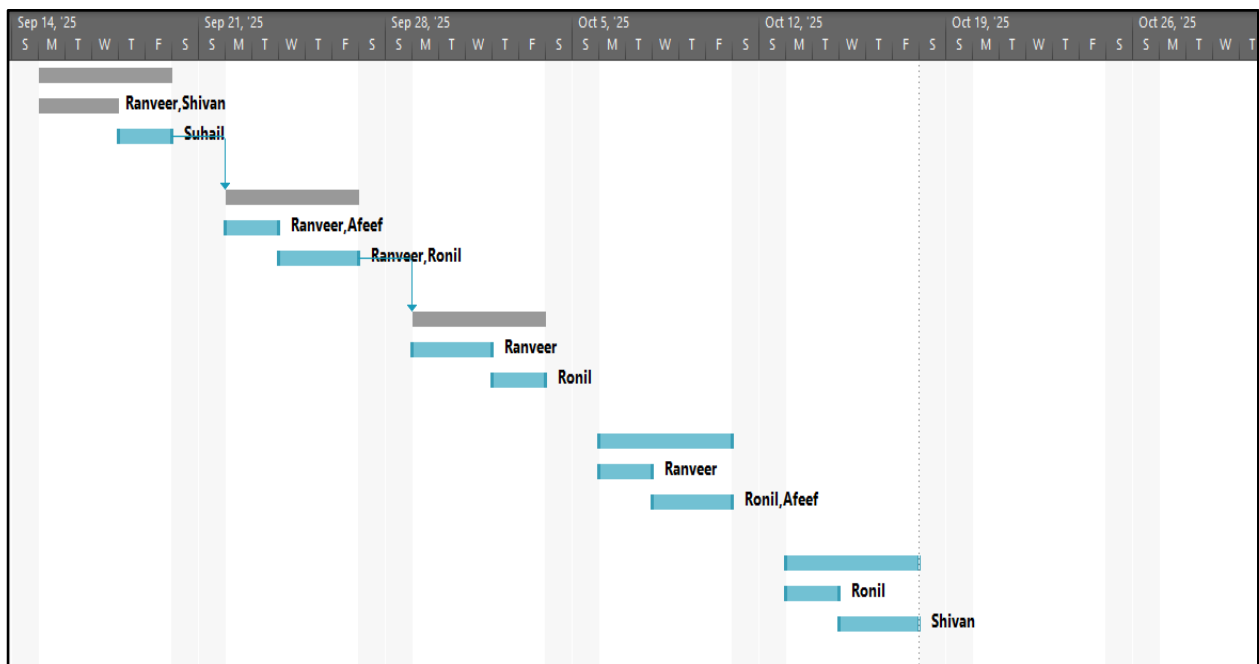
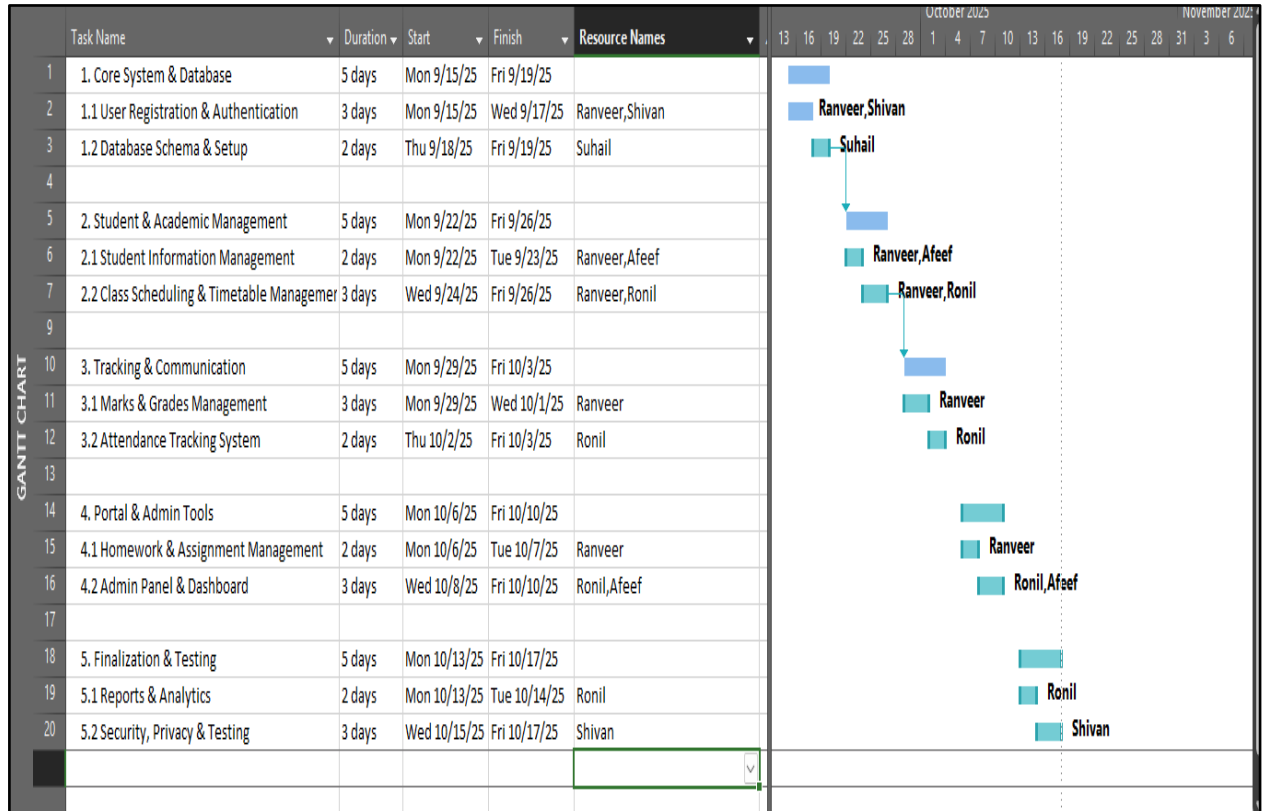
Week	Subsystem	Task Description	Assigned	Resources Required
Week 8	User Registration & Authentication	Create user roles (student, teacher, parent, admin staff, IT department), registration, and a login system with role-based access.	Shivan Prasad, Ranveer Singh	Authentication Libraries (JSON Web) Database (MySQL)
	Database Schema & Setup	Design the database schema to store the academic records, attendance, schedules, and user information.	Mohammed Suhail	DBMS (Database management System) and ERD (Entity Relationship Diagram)
Week 9	Student Information Management	Implement a comprehensive student profile, including enrollment history, personal details, and academic performance tracking.	Ranveer Singh, Mohammed Afeef	Database and UI/UX Tools (HTML, CSS, JavaScript)
	Class Scheduling & Timetable Management	Build a module for administrators to manage and allocate class schedules, for teachers or students to access the timetables.	Ranveer Singh, Ronil Prasad	Scheduling Algorithm and Calendar UI Component
Week 10	Marks & Grades Management	Develop some functionalities for teachers to enter grades, generate report cards, and track the academic progress of students.	Ranveer Singh	UI Forms and Database integration
	Attendance Tracking System	Implement a system for teachers to take	Ronil Prasad	Email/SMS Notification

		attendance and notify when students are absent.		Service, Attendance UI
Week 11	Homework & Assignment Management	Allows teachers to post assignments, and students to submit it digitally; set due dates and a feedback system.	Ranveer Singh	File Storage System (Cloud) and UI Tools
	Admin Panel & Dashboard	Create an admin dashboard to monitor the teachers' assignments, academic data, and system configurations.	Ronil Prasad, Mohammed Afeef	Admin Interface, Analytics Libraries
Week 12	Reports & Analytics	Implementing tools for tracking the attendance, teacher-student interactions, and academic performances.	Ronil Prasad	Reporting Tools (Charts, Graphs)
	Security, Privacy & Testing	Enforcing data encryption, privacy measures and conduct a security test to ensure compliance with data privacy standards.	Shivan Prasad	Encryption Libraries, Test Scripts

7.1 Gantt Chart

Student Administration System

Project Start: 12/09/2025



8.0 Definitions and Acronyms

SAS - Student administration system.

Scalability – Ability of the system to adapt to increased number of users.

Availability – The ability of the system to be available 24/7 even during the peak hours.

Mitigation – A risk management strategy used to minimize and avoid risks.

Data Flow Diagram – Shows flow of activities and data storage of the system.

Administrator - Is in charge of overseeing the system's general management.

DFD - Data Flow Diagram.

Functional Requirements - requirements that describe the features that a system must provide to meet user needs.

Non-Functional Requirements - define how a system should operate, rather than what it should do.

Implementation Plan - Is a comprehensive blueprint that describes the procedures needed to carry out a system or project from start to finish.

Module - An autonomous, self-contained entity that performs one or a set of tasks within a larger system.

Post-Condition - a condition that must be met following the execution of a procedure or function.

Gantt charts - Used to graphically depict a project's schedule, tasks, and timescale.

Structure chart - Using squares to symbolize the various system modules and lines to connect them, a structure chart is a top-down modular design tool.

9.0 References

- Boehm, B. (1989, September). Software risk management. In *European software engineering conference* (pp. 1-19). Berlin, Heidelberg: Springer Berlin Heidelberg.
- Geeksforgeeks (23 Aug, 2025) Functional and Non Functional Requirements .<https://www.geeksforgeeks.org/software-engineering/functional-vs-non-functional-requirements/>
- **User Interface** - Made with figma - <https://raven-vocal-07943442.figma.site/>

Assignment 1 Mark Allocation Sheet

After having discussed it as a group, we recommend the following mark allocation to each group member based on contribution or lack of it throughout the assignment.

Group number: 14

Project manager: Ranveer Singh

Member Student ID	Percentage Contribution	Responsibilities
S11230141	100%	Data flow diagrams & ToR Document
S11229568	100%	Use- case diagram & Implementation plan
S11230995	100%	ERD, Structured chart, Domain model class diagram & presentation
S11231541	100%	Introduction and purpose, Risk analysis, User interface & Definitions and Acronyms
S11231502	100%	System Capabilities and Scope, Sequence diagram & Design class diagram